

Microbiome

The human microbiome is composed of communities of bacteria, viruses and fungi that have a greater complexity than the human genome itself. Genome sequencing technologies and metagenomic analysis has helped in our understanding of human microbiome. This is useful in manipulation of gut microbiome to be used in the treatment of childhood diseases.

inserted into the enucleated cytoplasm of a host egg. The process results in the generation of an offspring that is genetically identical to the donor individual. The first animal produced by means of this cloning technique with a nucleus from an adult donor cell (as opposed to a donor embryo) was a sheep named Dolly, born in 1996. Since then a number of other animals, including pigs, horses, and dogs, have been generated by reproductive cloning technology. Recombinant DNA technology, on the other hand, involves the insertion of one or more individual genes from an organism of one species into the DNA (deoxyribonucleic acid) of another. Whole-genome replacement, involving the transplantation of one bacterial genome into the "cell body," or cytoplasm, of another microorganism, has been reported, although this technology is still limited to basic scientific applications.

Infobits

Gut Microbes

Human gut houses 1000 different species bacteria comprising of two major phyla namely Bacteroidetes and Firmicutes. The loss of balance in gut microbiota may lead to dybiosis. This may cause health problems like bowel disorders, bowel diseases like inflammation, allergies, obesity and diabetes.

1.5 Nanoparticles Production Using Microbes

Particles with one or more dimensions of the order of 100 nm or less. There are a large number of physical, chemical, biological, and hybrid methods available to synthesize different types of nanoparticles. Although physical and chemical methods are more popular in the synthesis of nanoparticles, the use of toxic chemicals greatly limits their biomedical applications, in particular in clinical fields. Therefore, development of reliable, nontoxic, and eco-friendly methods for synthesis of nanoparticles is of utmost importance to expand their biomedical applications. One of the options to achieve this goal is to use microorganisms to synthesize nanoparticles.

Nanoparticles are biosynthesized when the microorganisms grab target ions from their environment and then turn the metal ions into the element metal through enzymes generated by the cell activities. It can be classified into intra-cellular and extracellular synthesis according to the location where nanoparticles are formed. The intracellular method consists of transporting ions into the microbial cell to form nanoparticles in the presence of enzymes. The extracellular synthesis of nanoparticles involves trapping the

TN_GOVT_XII_Micro_Biology_CH01.indd 8 27-02-2019 15:14:49